

ANSWER KEY

SAT Practice Test 2 - Standardized Edition

Section 1: Reading and Writing

Module 1 | Module 2 Easier | Module 2 Harder

Section 2: Math

Module 1 | Module 2 Easier | Module 2 Harder

Answers and explanations are preserved from the original source file.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Question 1

Correct Answer: C) ongoing

The passage says Suzanne Dodd "continues to analyze transmissions," and the word must describe a contribution that is still happening. "Ongoing" means continuing, which directly matches the idea of a mission that has not ended.

Why other choices fail: "Unprecedented" (B) means never done before, which describes uniqueness, not continuity. "Debatable" (A) contradicts "confirming." "Peripheral" (D) means unimportant, which contradicts the positive tone.

Tip: When you see "continues" or "still" in the passage, the blank almost always needs a word conveying persistence or duration. Eliminate choices that describe quality (unprecedented) or value judgments (debatable, peripheral) when the passage emphasizes time.

Question 2

Correct Answer: A) He was violating an expectation about how to prepare for the tournament.

The narrator was "supposed to study" and was "meant to memorize" specific strategies, but instead he "skipped ahead" and moved pieces "wherever they seemed right." Improvising here means acting without following the prescribed preparation, thus violating the expectation that he would study properly.

Why other choices fail: "Gaining an unfair advantage" (B) is unsupported; improvising without studying would be a disadvantage. "Deceiving his chess instructor" (C) has no textual support; no instructor is mentioned. "Lying to himself" (D) is too strong; he simply did not study.

Tip: For "Words in Context" on literary passages, ground the meaning in what the character actually does, not what the word might mean in isolation. The narrator contrasts what he was told to do with what he actually did.

Question 3

Correct Answer: C) contributions

Volunteers "submit quotations illustrating word usage" from their own "personal libraries and archives." These submissions are voluntary offerings that help the dictionary. "Contributions" means something given to help a common effort, which fits perfectly.

Why other choices fail: "Resolutions" (B) means formal decisions. "Negotiations" (A) means discussions to reach agreement. "Justifications" (D) means reasons given to defend something. None describe voluntary submissions of materials.

Tip: Match the blank to the action described. Volunteers submit materials to aid a project. The word must mean "things given voluntarily." Only "contributions" captures that meaning.

Question 4

Correct Answer: D) expected by

The passage argues that polished acoustics are a modern development ("not widespread until the 1800s"). Therefore, centuries ago, audiences would not have had polished concert halls, and bare-walled venues would have been the norm. Something that is the norm is "expected."

Why other choices fail: "Confusing to" (B) implies audiences could not understand the experience, which is too strong. "Exciting to" (C) and "disliked by" (A) project emotions that the passage does not support.

Tip: Look for the logical contrast. The passage says modern audiences find bare walls startling because they are used to polished halls. Centuries ago the situation was reversed: bare walls were all audiences knew, so they would have expected them.

Question 5

Correct Answer: C) To suggest an unexpected solution to a problem

The text opens with "Struggling with an overpopulation of deer?" and then says "One surprising approach" is to reintroduce predators. The word "surprising" directly signals that the solution is unexpected. The entire passage is

built around proposing this idea.

Why other choices fail: "Discuss the best way" (B) is too broad; the text presents one approach, not a comprehensive discussion. "Describe how dangerous" (A) is wrong; the text focuses on helpfulness, not danger. "Identify a new ecological trend" (D) mischaracterizes the purpose.

Tip: Main purpose questions often hinge on the opening sentence. "Struggling with X? Here is a surprising approach" = suggesting an unexpected solution. Map the structure: problem then surprising solution.

Question 6

Correct Answer: A) It discusses fiction by earlier African American writers, then describes how fiction by more recent African American writers differs from the earlier fiction.

The first half discusses Morrison and her contemporaries (1970s and 1980s) and their "haunting, psychologically complex narratives." The second half introduces younger writers like Nnedi Okofor who have "gravitated toward" different genres like science fiction. This is a then vs. now structure.

Why other choices fail: "Praises then criticizes" (B) is wrong; neither writer is criticized. "Compares to non-Black writers" (C) is wrong; all writers discussed are Black. "Describes one author's early and recent works" (D) is wrong; two different authors are discussed.

Tip: For text structure questions, identify the pivot word. "Recently, however" signals a shift from earlier writers to newer ones and a contrast in genre.

Question 7

Correct Answer: A) It presents a criticism of the methodology reported by Dixson and colleagues in their 2016 study.

The underlined sentence says "Dixson's study relied on a sample of only 12 fish from a single reef system." The word "only" and the qualifier "a single" both signal that the sample is being characterized as limited, which is a methodological criticism.

Why other choices fail: "Lists traits" (B) is wrong; no traits of clownfish are listed. "Emphasizes where" (C) captures only part of the sentence; the emphasis is on the smallness of the sample, not just its location. "States the conclusion" (D) is wrong; the conclusion is in the first sentence.

Tip: When a sentence uses minimizing language ("only," "just," "merely," "a single"), it is usually functioning as a critique of scope or methodology. The follow-up about "skewed results" confirms this critical function.

Question 8

Correct Answer: A) Analogs allow scientists to test the ability of research equipment to withstand some of the conditions it will encounter during a mission.

The passage says analogs "help researchers evaluate how well their instruments will perform under Venusian conditions." This directly means analogs let scientists test equipment before sending it to Venus.

Why other choices fail: "Track chemical changes over time" (B) is not mentioned. "Combined with Earth materials" (C) distorts the passage. "Compare Venus to Earth" (D) is not the stated purpose.

Tip: For inference questions, find the sentence that most directly answers the question and paraphrase it. The passage says analogs help "evaluate how well instruments will perform," which matches choice A almost word for word.

Question 9

Correct Answer: D) 61.00 million metric tons of lithium in 2022.

The question says "Australia produced 5.20 million metric tons of lithium in 2000 and ____." Since the sentence is about Australia, we need Australia's 2022 figure from the table, which is 61.00.

Why other choices fail: Choice B (6.20) is Argentina's 2022 figure. Choice C (19.00) is China's 2022 figure. Choice A (39.00) is Chile's 2022 figure. All are from the correct year but the wrong country.

Tip: For data interpretation, read the stem carefully to identify which row and column you need. The sentence specifies Australia and pairs 2000 with a 2022 figure. Go directly to the Australia row, 2022 column.

Question 10

Correct Answer: B) In a Khmer stone inscription from approximately 683 CE, a dot appears between two numerals in a context that clearly indicates a positional value of nothing.

Keller's claim is that Khmer inscriptions from around 683 CE contain a dot functioning as a zero placeholder. Choice B provides direct evidence: a dot between numerals indicating "nothing," which is exactly what a zero placeholder does.

Why other choices fail: Choice A undermines the claim by suggesting dots had a different function. Choice C supports Brahmagupta, not the Khmer claim. Choice D says Coedes's readings are inaccurate, which would weaken the claim.

Tip: To strengthen a specific historical claim, you need evidence that directly matches the claim's specifics: the right date, the right artifact, and the right function. Choice B matches all three.

Question 11

Correct Answer: D) Sick days were fairly similar for the YGA and NTR groups at 6 months, but at 18 months they had significantly increased for the NTR group and significantly decreased for the YGA group.

The data shows YGA went from 3.1 to 1.8 (decreased) and NTR went from 3.3 to 4.9 (increased). At 6 months the values were close (3.1 vs 3.3), but at 18 months they diverged dramatically. Choice D accurately describes this pattern and supports the conclusion that yoga was more effective over time.

Why other choices fail: Choice A introduces "low morale" and "illness," which are not in the table. Choice B says YGA "barely changed," but 3.1 to 1.8 is a significant decrease. Choice C says the gap "narrowed," but it actually widened.

Tip: When a conclusion says an effect "took time to become apparent," look for data showing similar starting points but diverging endpoints. That pattern is what D describes.

Question 12

Correct Answer: C) "Cricket Flour Bar" and "Mealworm Pasta."

The statement says "the magnitude of this labeling effect varied." To best illustrate the largest magnitude of variation, you want the foods with the most dramatic labeling effect. Cricket Flour Bar and Mealworm Pasta have the largest gaps (roughly 3 points each), making them the most striking illustration of how much labels can influence ratings.

Why other choices fail: Choice A shows the smallest gaps. Choice B mixes a medium and large gap. Choice D mixes a medium and small gap. None illustrate the extreme end of the variation as dramatically as C.

Tip: When a graph question asks you to illustrate that an effect "varied," look for the data points that show the most extreme manifestation of that effect. The largest gaps demonstrate the upper range of the variation.

Question 13

Correct Answer: C) the greater exposure to humans among the captive lions induced them to acquire more right-pawed behaviors.

The passage notes that captive felines show more right-pawedness than wild felines. The logical completion should explain this discrepancy. Since captive animals have more human contact, and humans are predominantly right-handed, it is plausible that captive felines acquired right-pawed behaviors through exposure to humans.

Why other choices fail: Choices A and B discuss sample size, which does not explain the captive vs. wild pattern. Choice D suggests different criteria, which is possible but does not explain a systematic captive vs. wild difference.

Tip: For logical completion, identify what the passage is building toward. The key contrast is captive vs. wild. The blank must explain WHY captive felines show more right-pawedness. Human exposure is the most obvious difference between captive and wild animals.

Question 14

Correct Answer: A) there is seasonal flooding for far fewer than 90 days per year.

The passage says both species "prefer areas that experience seasonal flooding for more than 90 days per year." The conclusion is that "neither species is very drawn to areas where _____. Logically, if both prefer more than 90 days of flooding, neither would be attracted to areas with far fewer than 90 days.

Why other choices fail: Choice B introduces unrelated species. Choice C says "any features that attract the other," but the passage shows they share a preference. Choice D is wrong because only red-crowned cranes prefer shallow water, not

both species.

Tip: When a conclusion says "neither species is drawn to X," X must be the opposite of what both species prefer. Both prefer more than 90 days of flooding, so the opposite is far fewer than 90 days.

Question 15

Correct Answer: A) the edition of War and Peace has more words per page than the edition of The Goldfinch.

War and Peace: 580,000 words / 1,225 pages = approximately 474 words per page. The Goldfinch: 297,000 words / 771 pages = approximately 385 words per page. War and Peace has significantly more words per page.

Why other choices fail: Choice B makes an unsupported claim about translation. Choice C about reading time is not directly indicated by the data. Choice D about paper thickness is speculation without evidence.

Tip: When a logical completion question involves numbers, do the math. Divide word count by page count for each book and compare. The calculation is straightforward division.

Question 16

Correct Answer: B) Completed

"Completed around 1665, the painting can now be found..." requires a past participle to form a participial phrase modifying "the painting." "Completed" is the past participle that works as a modifier.

Why other choices fail: "Will complete" (A) is future tense and wrong direction. "Completes" (C) is present tense and would create a run-on. "Was completed" (D) would require the sentence to be restructured with a conjunction.

Tip: When a sentence starts with a blank followed by a time reference and then a comma, it usually needs a past participle to create a participial phrase. Test: "[Past participle] around [time], the [subject]..."

Question 17

Correct Answer: C) stems

The subject is "this capability" (singular), and we need a finite verb. "Stems" is a present-tense, singular verb that agrees with the singular subject and creates a complete clause: "this capability stems from..."

Why other choices fail: "Stemming" (A) and "having stemmed" (B) are participles that cannot serve as the main verb. "To stem" (D) is an infinitive that cannot serve as the main verb of the clause.

Tip: Identify the subject and check whether you need a finite verb or a modifier. "This capability" is the subject and needs a conjugated verb. "Stemming" and "having stemmed" are participles; "to stem" is an infinitive. Only "stems" is a finite verb.

Question 18

Correct Answer: B) languages—

The sentence contains a closing dash before "and accounts for." To create a properly paired parenthetical, you need an opening dash after "400 languages." This creates: "comprises more than 400 languages—Mandarin and Cantonese, which are spoken by...—and accounts for..."

Why other choices fail: "Languages" (A) with no punctuation creates an ungrammatical run ("400 languages Mandarin"). "Languages:" (C) introduces a list but does not pair with the closing dash. "Languages," (D) does not pair with the closing dash either.

Tip: When you see a closing dash later in a sentence, look for where the parenthetical begins and place a matching opening dash there. Paired dashes work like paired parentheses.

Question 19

Correct Answer: A) case; though,

Two independent clauses need proper separation. "This has not always been the case" is one clause; "throughout geologic history, the continents have collided..." is another. A semicolon properly separates them. "Though," set off by a comma, functions as a conjunctive adverb.

Why other choices fail: "Case, though," (B) creates a comma splice. "Case, though;" (C) has incorrect punctuation placement. "Case, though" (D) also creates a comma splice or run-on.

Tip: When you see two independent clauses, check for a semicolon or period. A comma alone (choices B, C, D patterns) would create a comma splice. The semicolon in A correctly separates the clauses.

Question 20

Correct Answer: C) rooms;

The sentence contains three independent clauses in a series: (1) "mirrored walls create infinite reflections," (2) "polka-dotted sculptures fill entire rooms," and (3) "suspended nets of light shift and shimmer." Since these are independent clauses, a semicolon is the correct separator.

Why other choices fail: "Rooms," (A) creates a comma splice between clauses 2 and 3. "Rooms—" (B) does not properly separate list items. "Rooms:" (D) implies what follows is an elaboration, but it is actually a third parallel item.

Tip: When you have a list of independent clauses (each with its own subject and verb), semicolons separate them, just as they would separate two independent clauses. Look for the pattern: [clause], [clause]; [clause].

Question 21

Correct Answer: B) Still,

The previous sentences acknowledge that Rococo ornamentation "couldn't exactly pass for a natural setting." Despite this concession, the next sentence says "one sees organic influences." "Still" signals a concession: despite what was just acknowledged, something is nevertheless true.

Why other choices fail: "Furthermore" (A) adds to the same point rather than introducing a counterpoint. "Similarly" (C) suggests parallel ideas, but the text contrasts ornament with nature. "In other words" (D) restates, but this is a new point.

Tip: "Granted... Still..." is a classic concession pattern. "Granted" acknowledges a point, and "Still" introduces a counterpoint. Look for this pattern on transition questions.

Question 22

Correct Answer: C) By comparison,

Japan requires voters to be 18; Scotland allows voting at 16. These are two different countries' voting ages being compared. "By comparison" introduces a contrasting example from a different context.

Why other choices fail: "Firstly" (A) implies a numbered list. "For example" (B) implies Scotland illustrates Japan's rule, which it does not. "Therefore" (D) implies Scotland's rule is a consequence of Japan's.

Tip: When two different entities or countries are presented with different policies or facts, "By comparison" is the correct transition to introduce the second one.

Question 23

Correct Answer: B) On the other hand,

The passage gives the Prius as an example of fuel efficiency, then presents a pickup truck as the opposite. "On the other hand" introduces a contrasting example.

Why other choices fail: "Specifically" (A) would elaborate on the Prius, not introduce a contrast. "As a result" (C) implies causation. "In conclusion" (D) signals a summary.

Tip: The structure is: general claim, then example supporting it, then example contrasting it. "On the other hand" bridges the supporting example to the contrasting one.

Question 24

Correct Answer: C) fittingly,

Channing "opposed slavery and advocated for human dignity." His sermons "challenged pro-slavery advocates." His actions match his beliefs. "Fittingly" means "in a way that is appropriate or suitable," connecting his abolitionist identity to his abolitionist sermons.

Why other choices fail: "In other words" (A) would restate the same point. "By comparison" (B) would introduce a contrast. "Nevertheless" (D) would introduce something contradictory to his beliefs.

Tip: When someone's actions are consistent with their stated beliefs or identity, "fittingly" is the correct transition. It means "as you would expect given what was just described."

Question 25

Correct Answer: A) The municipality of Tulum in Quintana Roo, Mexico, covers an area of roughly 2,090 km².

The goal is to emphasize the SIZE of Tulum. Only choice A mentions the specific area measurement (2,090 km²), which directly communicates size.

Why other choices fail: B discusses services, not size. C mentions the number of municipalities, not Tulum's size. D discusses waste management, unrelated to size.

Tip: For rhetorical synthesis, match the stated goal to the content of each choice. "Emphasize size" requires a size measurement. Scan for numbers that describe area, length, or magnitude.

Question 26

Correct Answer: A) Both the Gwaii Haanas NMCA and the Fathom Five NMCA were established to protect endangered species or ecosystems.

The goal is to emphasize a SIMILARITY. Both were established to protect endangered things (glass sponge reefs and shipwreck ecosystems). Choice A explicitly states "Both... were established to protect," highlighting the shared purpose.

Why other choices fail: B mentions only Fathom Five. C highlights a difference in size. D is about the total number of NMCAs, not a similarity between the two.

Tip: For "emphasize a similarity" questions, look for the choice that starts with "Both" or uses language that draws a parallel. Eliminate choices that highlight differences (size, location) or mention only one of the two.

Question 27

Correct Answer: A) Elements of classical Rinpa painting, manga, and corporate branding are playfully mixed in the Superflat piece Tan Tan Bo Puking by artist Takashi Murakami.

The goal is to "provide a specific example of playful mixing." Choice A names the specific work (Tan Tan Bo Puking) and lists the specific elements being mixed (Rinpa painting, manga, corporate branding). This is the most concrete, specific answer.

Why other choices fail: B quotes the artist but does not specify the work or the elements mixed. C names the work but does not list the elements. D mentions the work but describes the movement generally.

Tip: When the goal is "provide a specific example," the correct answer will name a particular work and describe particular elements. General statements about a movement are not specific examples.

Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Question 1

Correct Answer: B) dynamic

The passage defines the blank: "constantly shifting," "change in width and elevation." Something that is constantly shifting is dynamic. The colon after the blank signals that what follows is a definition.

Why other choices fail: "Unrivaled" (A) means unmatched, describing superiority. "Sustainable" (C) means maintainable, about longevity. "Immutable" (D) means unchangeable, which is the OPPOSITE of what is described.

Tip: When a colon follows the blank, the text after the colon defines the word. Read that definition and find the matching vocabulary word.

Question 2

Correct Answer: A) corroborated

Del Conde noticed a similarity and then "by digitally analyzing both works" she confirmed ("corroborated") the stylistic parallels. Corroborate means to confirm or support with evidence.

Why other choices fail: "Amalgamated" (B) means combined/merged. "Promulgated" (C) means made widely known. "Consolidated" (D) means combined into a single entity. None mean "confirmed."

Tip: When a scholar first argues something and then does analysis to confirm it, the verb must mean "confirmed" or "verified." "Corroborated" fits both blanks: Kahlo corroborated a shared vision through her art, and del Conde corroborated the parallels through digital analysis.

Question 3

Correct Answer: D) exhibit

Hadid's designs "can be said to _____ elements of both classical geometry and organic forms." The designs show or display these elements. "Exhibit" means to display or show.

Why other choices fail: "Defy" (A) means to resist or challenge. "Neglect" (B) means to fail to care for. "Foretell" (C) means to predict. None describe showing or displaying.

Tip: When the passage says designs "can be said to [blank] elements of X and Y," the verb needs to mean "show" or "display." Test each choice in the sentence.

Question 4

Correct Answer: B) mediated by

The data shows that the type of local policy framework (proactive vs. restrictive) was associated with very different likelihoods of establishing refugee programs (0.45 vs. 0.08). "Mediated by" means that the local policies served as the intermediary variable through which government responses operated.

Why other choices fail: "Predicated on" (A) means based on, which is close but suggests dependence rather than mediation. "Decoupled from" (C) means separated from, contradicting the data. "Material to" (D) means relevant to, which is too weak.

Tip: In social science contexts, "mediated by" means a variable acts as a mechanism or channel. The data shows local policies are that mechanism. "Predicated on" means "based on," which is close but less precise.

Question 5

Correct Answer: A) To present a reason why a literary critic is impressed by a certain novel

The text introduces Harold Bloom (a literary critic) and explains that he praises The Dispossessed for its "masterful synthesis" of utopian and critical modes. The text is presenting Bloom's reason for admiring the novel.

Why other choices fail: "Explain what inspired" (B) is wrong; Bloom's opinion, not Le Guin's inspiration, is the focus. "Compare writer and philosopher" (C) is wrong; Marcuse is used to frame Bloom's praise. "Argue all writing must be classified" (D) misrepresents the text; the point is that Le Guin transcended the classification.

Tip: Main purpose questions require you to identify WHO is doing WHAT. Bloom praises Le Guin's novel for a specific quality. That is the main purpose: presenting a critic's reason for being impressed.

Question 6

Correct Answer: B) It presents the favorable opinion of Mr. Turveydrop that other people hold and then describes the behaviors of Mr. Turveydrop that enable him to maintain that favorable opinion.

The first part describes how merchants and fellow dancing masters regard Turveydrop favorably. The second part describes his behaviors: never arguing with passion, never revealing his beliefs, never letting anyone detect he was performing a role. These behaviors explain HOW he maintains the favorable opinion.

Why other choices fail: Choice A is wrong; there is no naive vs. sophisticated distinction. Choice C is wrong; there is no decline in regard. Choice D is wrong; the text does not reveal his private behavior, only his public performance strategies.

Tip: For text structure in literary passages, separate what others think from what the character does. The first part is others' perception; the second part is his strategic behavior.

Question 7

Correct Answer: C) As a point of contrast with the Strand, Bond Street suggests to Clarissa the possibility of experiencing the lasting beauty promised by the contemplative spaces the novel seems to favor.

Text 2 says Bond Street "embodies an elegance that values beauty, permanence, and contemplation" as a contrast to the Strand's "noise, speed, and commercial energy." In Text 1, Clarissa sees Bond Street's stones as suggesting "permanence" and "a serenity beyond the grasp of commerce." Choice C captures this: Bond Street, as a contrast to the Strand, suggests lasting beauty and contemplation.

Why other choices fail: Choice A says Bond Street symbolizes commercial encroachment, which contradicts both texts. Choice B says Clarissa is dissatisfied, which is not supported. Choice D says Bond Street is "modest," but Text 1 describes it as dignified.

Tip: For dual-text questions, find the overlapping interpretation. Both texts describe Bond Street as a contrast to the Strand, associated with permanence and beauty rather than commerce.

Question 8

Correct Answer: D) It reflects an approach to form and shape that Niemeyer later moved away from.

The Santos Dumont Airport "reflect[s] the rectilinear aesthetics of traditional Brazilian Neoclassicism." This was Niemeyer's EARLIER style. After his European trip, he adopted "sweeping curves and open spaces." Thus, the airport reflects an approach he later moved away from.

Why other choices fail: Choice A says "transitional," but it is pre-transition. Choice B says it shows European influence, but it predates his European trip. Choice C says it influenced him throughout, but the passage says he moved to a different style.

Tip: Look for chronological markers. "Earlier projects" like the airport used rectilinear forms. His LATER work used curves. The airport represents the style he abandoned.

Question 9

Correct Answer: A) He has the appearance of a distinguished figure, but it is uncertain whether he has accomplished anything to earn distinction.

Boleyn's "expression seemed designed to ornament a monument" (looks distinguished), but examining "the actual fruits" of his service would be "discourteous" (implying the fruits are meager). His memoirs "might have illuminated this puzzle" but their substance was buried. He looks important but may have achieved nothing.

Why other choices fail: Choice B says "modest profile," but his expression is compared to a monument. Choice C says memoirs "reveal dishonorable actions," but the memoirs reveal nothing because their substance is buried. Choice D says achievements must be "kept secret," which is not supported.

Tip: In literary characterization, contrast appearance with substance. "Seemed designed to ornament a monument" describes impressive appearance. "Discourteous to examine the actual fruits" suggests the reality does not match.

Question 10

Correct Answer: D) The difference in average ISCH scores for songbird communities at Govan and Partick, two sites with a higher percentage of bank consisting of meadows than of wetlands, was statistically insignificant.

The finding is that meadows are more positively correlated with songbird health than wetlands. Choice D shows two meadow-dominant sites with consistent (statistically similar) ISCH scores, demonstrating that meadow-heavy sites reliably correlate with a particular level of songbird health.

Why other choices fail: Choice A contradicts the finding (high naturalization = lower scores). Choice B contradicts it (meadow site scores lower than wetland site). Choice C is ambiguous and does not isolate the meadow vs. wetland variable.

Tip: When "illustrating" a correlation, look for data that is consistent with, not contradictory to, the finding. A and B describe scenarios that contradict it. D is the only choice compatible with meadows being beneficial.

Question 11

Correct Answer: C) "The photograph I chose is accompanied by a statement of the personal interpretation I've developed over countless hours standing beside the work..."

The underlined claim says the show amplifies "a distinct perspective not typically given a formal platform." Choice C describes a guard's personal interpretation developed through "countless hours standing beside the work," shared previously only in "small impromptu conversations." This is a distinct perspective (from a guard's unique vantage point) not typically given a formal platform (previously shared only informally).

Why other choices fail: Choice A describes affection for the pieces but not a unique perspective or informal platform. Choice B is about lesser-known photographs, not about the guard's perspective. Choice D describes collaboration with professional curators, undermining the "distinct perspective" claim.

Tip: Match each element of the claim to the quotation. "Distinct perspective" = interpretation from hours of observation. "Not typically given a formal platform" = previously shared only in casual conversations.

Question 12

Correct Answer: C) "A growing population that is adding substantially more protein to its diet will contribute to an increasing demand for food production in the 21st century."

The student wants to support the claim that vertical farming may help address "an expected rise in global food demand." Choice C directly establishes that rising food demand is expected (growing population, more protein), which is the premise the student's claim rests on.

Why other choices fail: Choice A is about consumer beliefs on food safety. Choice B is about taste differences. Choice D describes vertical farming's environmental benefits but does not address rising food demand.

Tip: The claim has two parts: (1) food demand will rise, and (2) vertical farming can help. The quotation needs to support at least one of these. Choice C establishes part (1), giving the student a factual basis for the argument.

Question 13

Correct Answer: B) Many demonstrate the stylistic influence of bossa nova, a genre of Brazilian music...

The underlined claim says "the range of musical traditions woven into its fabric also expanded" as the movement spread beyond Argentina. Choice B shows a non-Argentine tradition (Brazilian bossa nova) being incorporated, directly illustrating the expansion of musical traditions.

Why other choices fail: Choice A involves shared South American experiences, not new musical traditions. Choice C is about the bandoneon, an Argentine instrument, not an expanded tradition. Choice D is about who produced the music, not about new musical traditions being incorporated.

Tip: The claim is about the range of traditions expanding as the movement spread geographically. You need a tradition from OUTSIDE Argentina being incorporated. Bossa nova is Brazilian, not Argentine.

Question 14

Correct Answer: D) the 2021 revision of the rule governing GCDs would allow expedited reviews of actions that had a minimal air-quality effect when considered on their own but a significant effect when considered together.

The old rule required that actions "do not individually OR CUMULATIVELY have a significant effect." The new rule only says "normally do not have a significant effect," dropping the cumulative requirement. This means actions that individually seem harmless but collectively cause significant damage could now get streamlined reviews.

Why other choices fail: Choice A is about incentives, which is not the textual focus. Choice B contradicts itself ("paradoxically slower"). Choice C reverses what happened (it says the old rule allowed expedited reviews for harmful actions).

Tip: Compare the old and new language word by word. The old rule said "individually or cumulatively"; the new rule dropped "cumulatively." That removal is what environmental groups found harmful.

Question 15

Correct Answer: D) graphene nanoparticles may allow electrons to bypass the series of metabolic steps and transfer directly to the circuit.

The passage says electrons normally pass through "inefficient metabolic steps." Graphene has "exceptional electrical conductivity." The output more than doubled with graphene. The hypothesis should logically connect graphene's conductivity to bypassing the inefficient steps.

Why other choices fail: Choice A says graphene increases metabolic processes, which contradicts "bypassing." Choice B suggests improvement without graphene. Choice C says electrons are conducted before graphene catalyzes, which is incoherent.

Tip: The hypothesis must explain WHY graphene increased output. Graphene conducts electricity well, so it likely provides a direct path for electrons, bypassing the inefficient metabolic steps.

Question 16

Correct Answer: D) memory, and

The sentence has two independent clauses: (1) "Her work challenges conventional notions of race, gender, history, and memory" and (2) "she is credited with expanding the horizons of textile-based art." Two independent clauses need a comma plus a coordinating conjunction ("and") to be properly joined.

Why other choices fail: "Memory," (A) creates a comma splice between two independent clauses. "Memory and" (B) creates "and memory and she," which is awkward. "Memory" (C) with no punctuation creates a run-on.

Tip: When you spot two independent clauses (each with subject + verb), check the options for a comma + coordinating conjunction ("and"). A comma alone creates a comma splice.

Question 17

Correct Answer: D) researchers, Odhiambo and Akech,

The phrase "according to legal researchers, Odhiambo and Akech," is a parenthetical. The opening comma is after "constitution." The closing comma must come after "Akech" to complete the parenthetical. "Odhiambo and Akech" is an appositive to "researchers," requiring a comma before and after.

Why other choices fail: Choice A has no closing comma after "Akech." Choice B has a closing comma but no comma between "researchers" and "Odhiambo." Choice C has a comma after "researchers" but no closing comma after "Akech."

Tip: When a parenthetical phrase with "according to" is embedded in a sentence, it needs commas at both ends. The comma after "Akech" closes the parenthetical that began after "constitution."

Question 18

Correct Answer: A) behavior. The

The sentence before the blank and the sentence after are both independent clauses. A period creates two separate, grammatically correct sentences. "They exhibit striking differences in behavior. The object 433 Eros is considered..." is clean and correct.

Why other choices fail: "Behavior; while the" (B) incorrectly combines a semicolon with "while." "Behavior, while the" (C) creates a comma splice. "Behavior; the" (D) works grammatically but the lowercase "the" creates an unusual construction.

Tip: When you have two independent clauses and the second one provides supporting detail, a period is the cleanest separator. Check that the resulting sentences each stand on their own.

Question 19

Correct Answer: D) determine

The subject is "a district's electoral history and polling data" (compound subject = plural). The phrase "not its population size" is a parenthetical and does not change the subject's number. A compound subject takes a plural verb: "determine."

Why other choices fail: "Has determined" (A) is singular and past. "Determines" (B) is singular. "Determining" (C) is a participle, not a finite verb. All fail to match the plural subject.

Tip: For subject-verb agreement with "X and Y, not Z, [verb]" constructions, the verb agrees with "X and Y" (plural), not with "Z." Strip out the parenthetical to see the true subject.

Question 20

Correct Answer: A) is due

The subject is "That the repeated eruptions... have occurred on timescales as brief as six months," which is a noun clause (starting with "That"). Noun clauses functioning as subjects take singular verbs. "Is due" is a singular finite verb.

Why other choices fail: "Due" (B) is not a verb; it would leave the sentence without a main verb. "Being due" (C) and "having been due" (D) are participles that cannot serve as the main verb.

Tip: When a sentence begins with "That [clause]," the entire "That..." clause is the subject and takes a singular verb. Strip away the modifiers to see: "That [X] is due to [Y]."

Question 21

Correct Answer: B) examining

The main verb is "reveals." The blank must be a present participle modifying "study," not a second main verb. "Examining" creates: "Hochschild's study [examining 60 interviews] reveals..."

Why other choices fail: "Had examined" (A) and "examines" (C) are finite verbs that would create a run-on with "reveals." "Examined" (D) is ambiguous and does not clearly modify "study."

Tip: When a sentence already has a main verb (here, "reveals"), the blank cannot be another finite verb. It must be a participle (modifier) or introduce a relative clause.

Question 22

Correct Answer: C) Consequently,

NASA crashed a probe into Dimorphos, and as a result, its orbital period changed. "Consequently" introduces a result or effect of the preceding action.

Why other choices fail: "In comparison" (A) introduces a contrast. "In addition" (B) adds unrelated information. "Specifically" (D) provides detail about the same point, not a result.

Tip: When event A (crashing the probe) directly causes event B (orbital period change), "Consequently" or "As a result" is the correct transition.

Question 23

Correct Answer: B) Rather,

The first sentence says metals "do not form bonds through directional sharing of electrons." The second sentence describes what they do instead: "held together by a sea of delocalized electrons." "Rather" introduces an alternative to what was just negated.

Why other choices fail: "For example" (A) would introduce an illustration, not a correction. "Accordingly" (C) implies a logical consequence. "Likewise" (D) implies similarity.

Tip: When a sentence says "X does NOT do Y," and the next sentence says "X does Z instead," use "Rather" to introduce the alternative.

Question 24

Correct Answer: C) likewise,

Both Keats and Smith are given as examples of ecphrastic poetry. Keats meditates on a vase; Smith takes Hubble photographs as her subject. Both are poets exploring visual art. "Likewise" connects two parallel examples.

Why other choices fail: "Nevertheless" (A) implies contrast. "Conversely" (B) implies opposition. "Hence" (D) implies causation.

Tip: When two examples serve the same illustrative purpose (both are ecphrastic poems), "likewise" or "similarly" is correct. The subjects differ (vase vs. space images) but the activity is the same.

Question 25

Correct Answer: D) That is,

The first sentence describes HbPLA's two functions (cleaving a substrate and acting as an antimicrobial peptide). The second sentence restates this by calling HbPLA "a bifunctional enzyme." "That is" introduces a restatement or clarification.

Why other choices fail: "Nevertheless" (A) implies contrast. "In fact" (B) intensifies but does not restate. "Moreover" (C) adds new information.

Tip: "That is" means "in other words" or "to put it another way." When the second sentence restates the first in different terms, "That is" is the correct transition.

Question 26

Correct Answer: B) National labor unions, such as the BGWF, work to improve conditions for workers in their member nations.

The goal is to "explain the purpose of national union federations." Choice B describes what national labor unions do (improve conditions for workers) and uses the BGWF as an example. This directly addresses purpose.

Why other choices fail: Choice A merely identifies the BGWF. Choice C describes a relationship with IndustriALL. Choice D describes international federations, not national ones.

Tip: The key word in the goal is "purpose." Look for the choice that explains WHAT national unions DO, not just what they are or who represents them.

Question 27

Correct Answer: B) Both PE and PTFE expand when heated, according to a 2020 study.

The goal is to "emphasize a similarity." Both PE and PTFE expanded when heated. Choice B states this shared behavior directly: "Both PE and PTFE expand when heated."

Why other choices fail: Choice A says "two research teams," which is factually incorrect (it was one study). Choice C highlights a difference (PTFE expanded more). Choice D is a general statement not specific to the two materials.

Tip: For "emphasize a similarity" questions, look for "Both X and Y [do the same thing]." Eliminate choices that highlight differences or mention only one material.

Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Question 1

Correct Answer: B) tenuous

The passage says attributions are often contradicted by chemical dating, "exposing a persistent gap between subjective expertise and empirical verification." Something that is frequently contradicted is "tenuous" (weak, not securely held).

Tip: Look for the word that means 'weak' or 'unreliable.' The passage describes attributions that are often overturned by evidence.

Question 2

Correct Answer: A) contingent on

The worm's metabolism is entirely dependent on the bacteria. It is 'incapable of independent nutrient acquisition.' 'Contingent on' means dependent on or conditional upon.

Tip: When an organism cannot function without something, its function is 'contingent on' that thing.

Question 3

Correct Answer: A) engender

Inclusive institutions produce sustained growth. 'Engender' means to cause or bring about. The passage contrasts inclusive institutions (which cause growth) with extractive ones (which cause poverty).

Tip: 'Engender' means to produce or generate. It is a positive causative verb that matches 'sustained growth.'

Question 4

Correct Answer: D) substantiated

The behavior 'has been _____ by decades of experimental results, leaving little room for alternative interpretations.' Decades of confirmation = substantiated (supported with evidence).

Tip: When evidence supports a claim over many years, it has been 'substantiated.' Look for words meaning 'confirmed by evidence.'

Question 5

Correct Answer: A) To explain why a scholar's unconventional methodology is suited to her subject matter

The text explains that Hartman blurs history and creative imagination because the traditional archive records the perspectives of those in power, not her subjects. The method matches the challenge.

Tip: The text describes what Hartman does (unconventional method), then explains why (the archive is biased). This structure explains why the method fits the subject.

Question 6

Correct Answer: D) It identifies an event and then presents three different hypotheses that attempt to explain it.

The text introduces the Cambrian Explosion, then presents three hypotheses: (1) rising oxygen, (2) predator-prey arms race, (3) Hox gene changes.

Tip: Count the hypotheses. 'Some paleontologists... Others, however... A third hypothesis...' = three distinct explanations for one event.

Question 7

Correct Answer: A) By noting that Chen's emphasis on listener awareness captures only one of several factors that influence code-switching decisions

Desai shows that topic and cultural association also drive code-switching, not just listener awareness. He would say Chen's factor is real but incomplete.

Tip: Desai does not contradict Chen; he adds another factor. This is a 'yes, but also' response, not a direct disagreement.

Question 8

Correct Answer: C) He is experiencing a conflict between the urgent need to act and his body's inability to comply.

Raskolnikov remembers the unlocked door ('urgent need to act') but 'his limbs refused to obey him' ('body's inability to comply'). He wants to leap up but cannot.

Tip: Look for the specific evidence. 'Would have leaped up, but his limbs refused' directly states the mind-body conflict.

Question 9

Correct Answer: A) ...natural light group reported an average improvement of 2.4 points...artificial light group reported an average improvement of 0.3 points.

The conclusion is that natural light has a GREATER EFFECT. Choice A shows the natural light improvement (2.4) vastly exceeding the artificial light improvement (0.3), directly supporting 'greater effect.'

Tip: To support 'X has a greater effect than Y,' show that X's measured outcome is significantly larger than Y's.

Question 10

Correct Answer: C) Countries with GDP per capita between \$10,000 and \$40,000 show a steady, linear increase in life expectancy with no sign of leveling off.

The claim is that the relationship is logarithmic (sharp increase then plateau). If the middle range shows linear increase with no plateau, the logarithmic pattern does not hold in that range.

Tip: To weaken a 'logarithmic' claim, find evidence of linearity where a plateau should be. Logarithmic patterns require leveling off at higher values.

Question 11

Correct Answer: C) Hydrological modeling shows that the Ghaggar-Hakra river system...changed course around 2000 BCE due to tectonic activity rather than reduced rainfall.

Vasquez claims reduced rainfall caused agricultural collapse. If the water loss was due to tectonic activity (river course change) rather than rainfall, her specific mechanism is wrong.

Tip: To weaken a causal hypothesis, show that the observed effect has a different cause. Vasquez says 'rainfall'; choice C says 'tectonic activity.'

Question 12

Correct Answer: B) "Arendt's portrait of Eichmann suggests that moral catastrophe can arise not from extraordinary malice but from the ordinary human tendency to follow orders without reflection."

The underlined claim says Eichmann's evil was 'thoughtlessness, an inability to consider consequences.' Choice B paraphrases this: 'ordinary tendency to follow orders without reflection.'

Tip: Match the quotation to the key terms in the claim. 'Thoughtlessness' = 'without reflection.' 'Not fanaticism' = 'not extraordinary malice.'

Question 13

Correct Answer: A) A Douglas fir seedling genetically related to a nearby mother tree received 40% more carbon through the fungal network than an unrelated seedling...

The claim says mother trees 'preferentially send carbon to their own seedlings.' Choice A directly shows a related seedling receiving more carbon than an unrelated one.

Tip: The key word is 'preferentially.' You need evidence comparing related vs. unrelated seedlings receiving different amounts.

Question 14

Correct Answer: B) employees are less likely to leave residual attention on a previous task if they have had the opportunity to bring it to completion before transitioning.

Leroy's finding is that finishing a task before starting a new one reduces attention residue. Uninterrupted work blocks let people finish tasks, which reduces residue, which improves productivity.

Tip: The blank must logically connect 'uninterrupted blocks' to 'improved productivity' through the mechanism of 'attention residue.' Choice B does this.

Question 15

Correct Answer: A) convection currents may play a role in some instances of the Mpemba effect but are not the sole mechanism responsible for the phenomenon.

The convection hypothesis predicts the effect should disappear without convection. Brownridge suppressed convection but still observed the effect. So convection is not the only cause, but may still contribute.

Tip: When an experiment removes variable X and the effect persists, X is not the sole cause. It may still be a contributing factor.

Question 16

Correct Answer: C) can regrow

The subject is 'the axolotl.' The sentence needs a finite verb phrase that describes what the axolotl does. 'Can regrow' is a modal verb + infinitive that creates a complete predicate.

Tip: Check that the blank produces a complete clause. 'To regrow' and 'regrowing' are non-finite. 'Having regrown' is a perfect participle. Only 'can regrow' works as a main verb.

Question 17

Correct Answer: D) had predicted

The galaxies are 'far more luminous and massive than existing models _____. The models' predictions came before the telescope's observations. Past perfect ('had predicted') correctly places the prediction before the discovery.

Tip: Use past perfect when one past event (prediction) preceded another past event (discovery). Models had predicted X before the telescope revealed Y.

Question 18

Correct Answer: A) but also the work of

The sentence uses 'not only... but also...' construction. 'Not only traditional art forms... but also the work of contemporary architects.' No extra commas needed in this construction.

Tip: The 'not only... but also...' correlative conjunction must be completed with 'but also' and no intervening comma.

Question 19

Correct Answer: A) itself,

The sentence has a main clause ('Fleming noticed that a mold colony had created a bacteria-free zone around itself') followed by an appositive ('an observation that led to...'). A comma separates the main clause from the appositive.

Tip: When a noun phrase restates or elaborates on the preceding clause, a comma introduces the appositive. 'Around itself, an observation that...'

Question 20

Correct Answer: B) is

The subject is 'The collection' (singular). 'Along with the accompanying archive' is a prepositional phrase that does not change the subject's number. Singular subject takes singular verb: 'is.'

Tip: 'Along with,' 'together with,' and 'as well as' do not create compound subjects. The verb agrees with the first noun only.

Question 21

Correct Answer: D) For instance,

Lakoff argues that abstract thinking uses metaphor. The example of 'approaching deadline' and 'past is behind' illustrates this claim. 'For instance' introduces a specific example.

Tip: When a general claim is followed by a specific illustration, 'For instance' or 'For example' is the correct transition.

Question 22

Correct Answer: C) In contrast,

Germany shut down nuclear reactors; France is building more. These are opposite responses to the same issue. 'In contrast' introduces an opposing action.

Tip: When two entities take opposite approaches to the same situation, 'In contrast' signals the shift.

Question 23

Correct Answer: D) evidently,

The researchers found that reading literary fiction improves empathy scores. The skill of perspective-taking 'evidently transfers' to real-world interactions. 'Evidently' means 'as the evidence suggests.'

Tip: 'Evidently' means 'as shown by the evidence.' It connects the research finding to the real-world application.

Question 24

Correct Answer: C) In particular,

Kahneman showed people deviate from rationality in predictable ways. Loss aversion is a specific example of such a deviation. 'In particular' introduces a specific instance of a general claim.

Tip: 'In particular' narrows from a general statement to a specific example. Kahneman showed deviations generally; loss aversion is one particular example.

Question 25

Correct Answer: B) Although the Ocean Cleanup project has developed systems to capture floating debris from the GPGP, 94% of the patch consists of microplastics dispersed throughout the water column, making comprehensive removal extremely difficult.

The goal is to explain WHY cleanup is difficult DESPITE technology. Choice B contrasts the technology (captures floating debris) with the problem (94% are dispersed microplastics), explaining the difficulty.

Tip: For 'explain why X despite Y,' the answer must contain both the effort (Y) and the obstacle (X), connected logically.

Question 26

Correct Answer: B) While CRISPR-Cas9 holds promise for improving food security and treating genetic disorders like sickle cell disease, its potential use for germline editing has raised ethical concerns about heritable changes to the human genome.

The goal is to 'highlight a contrast between benefits and ethical concerns.' Choice B does both: benefits (food security, treating disorders) on one side, ethical concerns (germline editing) on the other.

Tip: For 'contrast X and Y,' the answer must contain both X and Y with a contrasting connector like 'while' or 'although.'

Question 27

Correct Answer: C) The vault was designed to store backup copies of seeds so that lost collections could be replaced, and in 2015, seeds were withdrawn for the first time to restore gene bank collections destroyed during the Syrian civil war.

The goal is 'evidence that the vault has served its intended purpose.' Its purpose is to provide backup seeds when collections are lost. In 2015, seeds were withdrawn to replace collections destroyed in Syria. This is the vault doing exactly what it was designed to do.

Tip: For 'evidence it served its purpose,' match the stated design goal to an actual instance of that goal being fulfilled.

Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Question 1

Correct Answer: C) 40

$d = 25 + 3t$. Substitute $t = 5$: $d = 25 + 3(5) = 25 + 15 = 40$.

Tip: Plug in the given value directly. This is a simple substitution question.

Question 2

Correct Answer: C) $7^2 + a^2 = 25^2$

By the Pythagorean theorem, in a right triangle: $(\text{leg})^2 + (\text{leg})^2 = (\text{hypotenuse})^2$. So $7^2 + a^2 = 25^2$.

Tip: Identify the hypotenuse (the longest side, opposite the right angle) and apply $a^2 + b^2 = c^2$.

Question 3

Correct Answer: D) 47

The last bar covers weights from 40 to less than 50 kg and has 7 data points. So the maximum weight must be in this range. 47 is less than 50, so it could be the maximum.

Tip: The maximum must fall within the highest bar's range. Eliminate values outside that range. 9 is in bar 1, 29 in bar 3, 39 in bar 4.

Question 4

Correct Answer: B) 9

$3x + 12 = 27$. Notice that $3x + 12 = 3(x + 4)$. So $3(x + 4) = 27$, giving $x + 4 = 9$.

Tip: Factor the left side to match the expression asked for. $3x + 12 = 3(x + 4)$, so divide both sides by 3.

Question 5

Correct Answer: 35

For infinitely many solutions, both sides must be identical. $9x + 35 = 9x + k$ requires $k = 35$.

Tip: Infinitely many solutions means the equation is an identity. The constants on both sides must match.

Question 6

Correct Answer: 52

$156 = 52 + 2p$, where p is each equal piece. $2p = 104$, so $p = 52$.

Tip: Total = known piece + 2(unknown piece). Solve for the unknown piece.

Question 7

Correct Answer: 6

Let n = notebooks, p = pens. $n + p = 15$ and $4n + 2p = 42$. From the first equation, $p = 15 - n$. Substitute: $4n + 2(15 - n) = 42$. Simplify: $4n + 30 - 2n = 42$, so $2n = 12$, $n = 6$.

Tip: Set up a system of equations. Substitution is efficient here since one equation can be easily solved for one variable.

Question 8

Correct Answer: D) $y = -x + 4.0$

The line passes through $(-12, 16)$ and $(0, 4)$. Slope = $(4 - 16)/(0 - (-12)) = -12/12 = -1$. Y-intercept = 4. So $y = -x + 4$.

Tip: Use two points on the line to calculate slope: $(y_2 - y_1)/(x_2 - x_1)$. Then use the y-intercept.

Question 9

Correct Answer: 7

$5x^4 + 11x^4 - 9x^4 = (5 + 11 - 9)x^4 = 7x^4$. So $b = 7$.

Tip: Combine like terms by adding/subtracting the coefficients.

Question 10

Correct Answer: C) (8, 29)

Set $3x + 5 = 4x - 3$. Solving: $8 = x$. Then $y = 3(8) + 5 = 29$. The point is $(8, 29)$.

Tip: Set the two equations equal to find x , then substitute back to find y .

Question 11

Correct Answer: A) The length of the rectangle, in meters

Length = 45 times width = $45w$. The function $y = (45w)(w)$ means area = length times width. So $45w$ represents the length.

Tip: In area = length times width, identify which factor is which. Since length = 45 times width, $45w$ is the length.

Question 12

Correct Answer: B) x : 0, 1, 2 $\rightarrow$ $f(x)$: 37, 37, 37

$f(x) = 37$ is a constant function. For ANY value of x , $f(x) = 37$. So $f(0) = f(1) = f(2) = 37$.

Tip: A constant function outputs the same value regardless of input. Every x gives 37.

Question 13

Correct Answer: B) $y = 145$

If $w = 145$ and $y = 145$, then w and y are equal. If w and y are corresponding angles (or alternate interior angles), then equal values prove lines r and s are parallel.

Tip: For parallel lines, you need corresponding angles equal, alternate angles equal, or co-interior angles summing to 180. $y = w = 145$ establishes corresponding angles equal.

Question 14

Correct Answer: C) 80

Complete the square: $(x - 3)^2 + (y - 5)^2 = 9 + 25 + 66 = 100$. Radius = 10. Diameter = 20. Square side = 20. Perimeter = $4(20) = 80$.

Tip: Complete the square for both x and y terms. The radius gives you the diameter, which equals the side of the circumscribed square.

Question 15

Correct Answer: 176

$800 \times 0.55 = 440$ attended first and second. $440 \times 0.40 = 176$ attended all three.

Tip: Apply successive percentages: 55% of 800, then 40% of that result.

Question 16

Correct Answer: B) x : -5, 0, 5, 10 $\rightarrow$ $f(x)$: 10, 30, 90, 270

$f(-5) = 30(3^{-1}) = 10$. $f(0) = 30(3^0) = 30$. $f(5) = 30(3^1) = 90$. $f(10) = 30(3^2) = 270$.

Tip: Substitute each x value. Remember: $3^{-1} = 1/3$, $3^0 = 1$, $3^1 = 3$, and $3^2 = 9$.

Question 17

Correct Answer: 471

$V = \pi r^2 h = \pi(25)(8) = 200\pi$. 75% of $200\pi = 150\pi \approx 150(3.14) = 471$.

Tip: Calculate full volume first, then take 75% of it. Use $\pi \approx 3.14$ as directed.

Question 18

Correct Answer: 32

Volume = edge³ = 4³ = 64 cm³. Mass = density × volume = 0.500 × 64 = 32 grams.

Tip: For cubes: $V = s^3$. Then mass = density × volume.

Question 19

Correct Answer: B) 0 and -3

$-3|4x + 6| + 9 = -9$. Solve: $-3|4x + 6| = -18$, so $|4x + 6| = 6$. Two cases: $4x + 6 = 6$ gives $x = 0$. $4x + 6 = -6$ gives $x = -3$.

Tip: Isolate the absolute value first. Then split into two cases: positive and negative.

Question 20

Correct Answer: D) 72

By Vieta's formulas, for $ax^2 + bx + c = 0$, the sum of solutions = $-b/a = -(-72)/1 = 72$.

Tip: Use Vieta's formulas: sum of roots = $-b/a$. No need to solve the quadratic.

Question 21

Correct Answer: 1024

Slant height = 20, height = 12. Radius = $\sqrt{20^2 - 12^2} = \sqrt{400 - 144} = \sqrt{256} = 16$. Volume = $(1/3)\pi r^2 h = (1/3)\pi(256)(12) = 1024\pi$. So $k = 1024$.

Tip: Use the Pythagorean theorem with slant height and height to find the radius. Then apply the cone volume formula.

Question 22

Correct Answer: A) $(5, 6 - 1/m)$

Line j has slope m . Perpendicular line m has slope $-1/m$. Line m passes through $(4, 6)$. At $x = 5$: $y = 6 + (-1/m)(5 - 4) = 6 - 1/m$.

Tip: Perpendicular slopes are negative reciprocals. Use point-slope form with the intersection point.

Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

Question 1

Correct Answer: A) 1.2 seconds after being launched, the ball's height above ground is 4.5 meters.

The point (1.2, 4.5) means $x = 1.2$ (seconds) and $y = 4.5$ (meters). The x-axis is time and the y-axis is height.

Tip: For graph interpretation, always check what each axis represents. $(x, y) = (\text{time}, \text{height})$.

Question 2

Correct Answer: C) The amount, in dollars, the freelancer earned for each hour at the primary contract

In $20h + 15c = 900$, the coefficient 20 multiplies h (hours at primary contract). Since the equation gives total dollars, 20 must be the hourly rate at the primary contract.

Tip: Coefficients in linear equations represent rates. The coefficient of h is the rate per hour at the primary contract.

Question 3

Correct Answer: D) 32, 33, 33, 33, 34

Standard deviation measures spread. The set {32, 33, 33, 33, 34} is the most tightly clustered around its mean (33), so it has the smallest standard deviation.

Tip: For standard deviation comparisons, look for the data set where values are closest to each other. No calculation needed; visual inspection works.

Question 4

Correct Answer: B) 62°

Similar triangles have equal corresponding angles. Angle FGH corresponds to angle FIJ = 62° . Side ratios do not affect angles.

Tip: In similar triangles, corresponding angles are always equal regardless of size. The side ratio $FH = 18(IJ)$ is irrelevant to the angle measure.

Question 5

Correct Answer: D) 50

$P(Z) = 1 - P(X) - P(Y) = 1 - 0.54 - 0.36 = 0.10$. Number in Z = $0.10 \times 500 = 50$.

Tip: Probabilities must sum to 1. Subtract the given probabilities to find the remaining one, then multiply by the total.

Question 6

Correct Answer: A) 24

Set equal: $x/3 + 8 = -x/3 + 16$. Solving: $2x/3 = 8$, so $x = 12$. $y = 12/3 + 8 = 12$. $2y = 24$.

Tip: Set the two expressions equal, solve for x , then find y , and finally compute $2y$.

Question 7

Correct Answer: B) $y = 31.5$

The total amount consumed in 7 days was 31.5 grams. y represents this total, so $y = 31.5$.

Tip: Read carefully: y is the total amount consumed in 7 days, not the daily rate. The total is 31.5.

Question 8

Correct Answer: D) $M = 50(0.82)^{t/8.5}$

Decreasing by 18% means retaining 82% = 0.82 each period. The period is 8.5 minutes, so the exponent is $t/8.5$.

Tip: Decay by $r\%$ means multiply by $(1 - r/100)$ each period. The exponent is t divided by the period length.

Question 9

Correct Answer: D) $y = -5x + 12$

Parallel lines have the same slope. Slope = -5 . Passes through $(0, 12)$, so y-intercept = 12 . Equation: $y = -5x + 12$.

Tip: Parallel = same slope. The new y-intercept is the y-coordinate of the given point when the point has $x = 0$.

Question 10

Correct Answer: C) 6

g has slope 6 . $g(c) = -4$ and $g(3) = 32$. So $32 - (-4) = 36 = 6(3 - c)$, giving $c = -3$. For h : $h(-3) = -5$ and $h(6) = 49$. Slope = $(49 - (-5))/(6 - (-3)) = 54/9 = 6$.

Tip: First find c using g 's slope. Then use c to compute h 's slope from two known points.

Question 11

Correct Answer: -1.5

Let $u = 5 - 2x$. Then $3u + 8u + 4 = 7u + 10$, so $11u + 4 = 7u + 10$, giving $4u = 6$, $u = 3/2$. Since $u = 5 - 2x$, we get $2x - 5 = -u = -3/2 = -1.5$.

Tip: Use substitution: let $u = 5 - 2x$ to simplify, then note $2x - 5 = -u$.

Question 12

Correct Answer: D) 6

From point $(0, -6)$: $r(0) + t(-6) = -48$, so $t = 8$. From point $(-8, -8)$: $r(-8) + 8(-8) = -48$, so $-8r - 64 = -48$, giving $r = -2$. Therefore $r + t = -2 + 8 = 6$.

Tip: Substitute two points on the boundary line into the equation $rx + ty = -48$ to create a system and solve for r and t .

Question 13

Correct Answer: D) $-1/3$

$f(a) = -3a$ means $\sqrt{(3a + 2)} = -3a$. Square both sides: $3a + 2 = 9a^2$, so $9a^2 - 3a - 2 = 0$. Factor: $(3a + 1)(3a - 2) = 0$, giving $a = -1/3$ or $a = 2/3$. Check: $a = -1/3$ gives $\sqrt{1} = 1$ and $-3(-1/3) = 1$. Valid. $a = 2/3$ gives $\sqrt{4} = 2$ but $-3(2/3) = -2$. Invalid since $\sqrt{4} \neq -2$.

Tip: After solving, always check both solutions in the ORIGINAL equation. Square root equations often produce extraneous solutions.

Question 14

Correct Answer: $5/4$

The fourth root of p^6 is $p^{6/4} = p^{3/2}$. If $t = p^{(2n - 1)}$, then $3/2 = 2n - 1$, so $n = 5/4$.

Tip: Convert the radical to a fractional exponent, then set exponents equal since the bases are the same.

Question 15

Correct Answer: C) 272

Vertex form: $g(x) = a(x - 1)^2 + 4$. Using $(2, 38)$: $a(1) + 4 = 38$, so $a = 34$. $g(-2) = 34(9) + 4 = 310$. $g(0) = 34(1) + 4 = 38$. $g(-2) - g(0) = 310 - 38 = 272$.

Tip: Use vertex form $g(x) = a(x - h)^2 + k$ with the vertex. Plug in a known point to find a .

Question 16

Correct Answer: 42

$f(0) = -2(0 - 3)(0 + 7) = -2(-3)(7) = -2(-21) = 42$.

Tip: The y-intercept is $f(0)$. Substitute $x = 0$ into the factored form.

Question 17

Correct Answer: B) $f(x) = -4^x + 3$

The graph of $y = f(x) + 3$ passes through (0, 5) and approaches 6 from below. So $f(x) + 3 = -4^x + 6$. At $x = 0$: $-1 + 6 = 5$. Correct. Therefore $f(x) = -4^x + 3$.

Tip: Read the graph's behavior: horizontal asymptote and key points. Then subtract 3 to find $f(x)$.

Question 18

Correct Answer: C) 22.86

Original: 60 inches per side, 1 inch = 8 miles. Total actual side = $60 \times 8 = 480$ miles. Smaller: side = $60 \times 0.35 = 21$ inches. New scale: $480/21 \approx 22.86$ miles per inch.

Tip: The actual area does not change; only the blueprint size changes. Divide the actual distance by the new number of inches.

Question 19

Correct Answer: 7/2

Solutions: $x = 0$, $x = 5/2$, $x = u/2$. Sum: $0 + 5/2 + u/2 = 17/4$. So $(5 + u)/2 = 17/4$, giving $u = 7/2$.

Tip: Set each factor to zero to find solutions. Then set their sum equal to $17/4$ and solve for u .

Question 20

Correct Answer: 2520

Scale factor k : $k^2 = 1800/72 = 25$, so $k = 5$. Volume of $P = 2500/k^3 = 2500/125 = 20$. Sum = $20 + 2500 = 2520$.

Tip: For similar figures: surface area ratio = k^2 , volume ratio = k^3 . Find k from surface areas, then use it for volumes.

Question 21

Correct Answer: 5832

$g(6)/g(3) = d^{(3/m)} = 216/8 = 27$. $g(9) = g(6) \times d^{(3/m)} = 216 \times 27 = 5832$.

Tip: In exponential functions with equal x -intervals, the ratio between consecutive outputs is constant. Use this multiplier pattern.

Question 22

Correct Answer: D) 1.25

$h = -16t^2 + 40t + 6$. Maximum at $t = -b/(2a) = -40/(2(-16)) = 40/32 = 1.25$.

Tip: For a quadratic $h = at^2 + bt + c$ with $a < 0$, the maximum occurs at $t = -b/(2a)$.

Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Question 1

Correct Answer: B) 5

$2x + 3y = 14$ and $5x + 7y = 33$. Multiply equation 1 by 7: $14x + 21y = 98$. Multiply equation 2 by 3: $15x + 21y = 99$. Subtract: $x = 1$. Then $2(1) + 3y = 14$, so $3y = 12$, $y = 4$. Therefore $x + y = 1 + 4 = 5$.

Tip: Use elimination to solve systems. Multiply equations to make one variable's coefficients equal, then subtract.

Question 2

Correct Answer: 7.5

Let a = liters of Solution A, b = liters of Solution B. $a + b = 20$ and $0.30a + 0.70b = 0.55(20) = 11$. From the first: $b = 20 - a$. Substitute: $0.30a + 0.70(20 - a) = 11$. $0.30a + 14 - 0.70a = 11$. $-0.40a = -3$. $a = 7.5$.

Tip: Set up a mixture equation: (concentration 1)(amount 1) + (concentration 2)(amount 2) = (desired concentration)(total amount).

Question 3

Correct Answer: 5

$4(3x + k) = 12x + 4k$. For this to equal $12x + 20$, we need $4k = 20$, so $k = 5$.

Tip: Distribute the left side and compare coefficients. For infinitely many solutions, both sides must be identical.

Question 4

Correct Answer: A) 11

Vertex form: $f(x) = 2(x - 3)^2 + (-7) = 2(x - 3)^2 - 7$. $f(0) = 2(0 - 3)^2 - 7 = 2(9) - 7 = 18 - 7 = 11$.

Tip: Write the quadratic in vertex form using the given vertex, then evaluate at $x = 0$.

Question 5

Correct Answer: -5

$f(x) = (x + 3)^2 - 16$ has its minimum at $x = -3$. $g(x) = f(x + 2)$ shifts the graph left by 2, so the minimum is at $x = -3 - 2 = -5$.

Tip: $g(x) = f(x + 2)$ shifts the graph of f LEFT by 2 units. Shift the vertex x -coordinate accordingly.

Question 6

Correct Answer: C) 8

$h(h(x)) = a(ax + b) + b = a^2x + ab + b = 9x + 20$. So $a^2 = 9$, giving $a = 3$. Then $ab + b = 3(5) + 5 = 20$, so $b = 5$. Check: $h(3x+5) = 3(3x+5) + 5 = 9x + 20$. Correct. $a + b = 3 + 5 = 8$.

Tip: Compose h with itself: $h(h(x)) = a(ax+b)+b = a^2x + ab + b$. Set coefficients equal to $9x + 20$.

Question 7

Correct Answer: D) 29

$(x + y)^2 = x^2 + 2xy + y^2 = 49$. Since $xy = 10$, we have $2xy = 20$. So $x^2 + y^2 = 49 - 20 = 29$.

Tip: Use the identity $(x + y)^2 = x^2 + 2xy + y^2$. Subtract $2xy$ to isolate $x^2 + y^2$.

Question 8

Correct Answer: 12/5

$27^x = 9^{y+1}$. Rewrite: $3^{3x} = 3^{2(y+1)} = 3^{2y+2}$. So $3x = 2y + 2$. With $x + y = 5$ (so $y = 5 - x$): $3x = 2(5 - x) + 2 = 12 - 2x$. $5x = 12$. $x = 12/5$.

Tip: Convert to the same base (base 3). Then set exponents equal and solve the system.

Question 9

Correct Answer: -2

By the Remainder Theorem: $p(1) = 2 + a + b + 6 = 3$, so $a + b = -5$. $p(-1) = -2 + a - b + 6 = 5$, so $a - b = 1$. Adding the two equations: $2a = -4$, giving $a = -2$. (Then $b = -3$.)

Tip: Use the Remainder Theorem: $p(c)$ equals the remainder when $p(x)$ is divided by $(x - c)$. Set up two equations and solve.

Question 10

Correct Answer: B) 12

For exactly one solution, the discriminant must be zero: $b^2 - 4ac = k^2 - 4(1)(36) = k^2 - 144 = 0$. So $k^2 = 144$, $k = 12$ (since k is positive).

Tip: One solution (repeated root) means discriminant = 0. Set $b^2 - 4ac = 0$ and solve for k .

Question 11

Correct Answer: 9

$f(x) = -x^2 + 10x - 16$. Maximum at $x = -b/(2a) = -10/(2(-1)) = 5$. $f(5) = -25 + 50 - 16 = 9$.

Tip: Find the vertex x -coordinate with $-b/(2a)$, then evaluate f at that point.

Question 12

Correct Answer: D) 4

Let the numbers be p and q . $p + q = 20$ and $pq = 96$. $(p - q)^2 = (p + q)^2 - 4pq = 400 - 384 = 16$. $|p - q| = 4$. (The numbers are 8 and 12.)

Tip: Use the identity $(p - q)^2 = (p + q)^2 - 4pq$ to find the difference without finding individual values.

Question 13

Correct Answer: 60/13

$AC = \sqrt{(13^2 - 5^2)} = \sqrt{(169 - 25)} = \sqrt{144} = 12$. Area of triangle $ABC = (1/2)(5)(12) = 30$. Also area = $(1/2)(BC)(BD) = (1/2)(13)(BD)$. So $30 = (13/2)(BD)$, giving $BD = 60/13$.

Tip: Calculate the area two ways: using the two legs, and using the hypotenuse with the altitude. Set them equal to find the altitude.

Question 14

Correct Answer: B) 18

From points $(2,6)$ and $(8,6)$, the center's x -coordinate is $(2+8)/2 = 5$. From points $(2,6)$ and $(2,0)$, the center's y -coordinate is $(6+0)/2 = 3$. Center = $(5,3)$. $r^2 = (5-2)^2 + (3-0)^2 = 9 + 9 = 18$. Verify with all three points: each gives $r^2 = 18$.

Tip: Find the center using perpendicular bisectors. The center is equidistant from all three points.

Question 15

Correct Answer: 504 π

Cylinder volume = $\pi(6)^2(10) = 360\pi$. Hemisphere volume = $(2/3)\pi(6)^3 = (2/3)(216)\pi = 144\pi$. Total = $360\pi + 144\pi = 504\pi$.

Tip: Add the cylinder volume ($\pi r^2 h$) and the hemisphere volume $((2/3)\pi r^3$, which is half of $(4/3)\pi r^3$.

Question 16

Correct Answer: C) 24.5

Line through $(2,5)$ and $(6,1)$: slope = $(1-5)/(6-2) = -1$. Equation: $y - 5 = -1(x - 2)$, so $y = -x + 7$. x -intercept: $0 = -x + 7$, $x = 7$, point $(7,0)$. y -intercept: $x = 0$, $y = 7$, point $(0,7)$. Triangle area = $(1/2)(7)(7) = 24.5$.

Tip: Find the line equation, then find x and y intercepts. The triangle has legs along the axes; area = $(1/2)(\text{base})(\text{height})$.

Question 17

Correct Answer: A) $y = -(x - 3)^2 + 2$

Reflecting $y = (x-3)^2 + 2$ over the line $y = 2$: the vertex (3,2) is ON the line $y = 2$, so it stays fixed. The parabola opens upward; reflected, it opens downward. Result: $y = -(x-3)^2 + 2$.

Tip: When reflecting over $y = k$, points on the line stay fixed, and the direction of opening reverses. If the vertex is on the line of reflection, only the sign of the leading coefficient changes.

Question 18

Correct Answer: B) 27

Original sum = $10 \times 24 = 240$. New sum = $240 + 57 = 297$. New mean = $297/11 = 27$.

Tip: New mean = (old sum + new value) / (old count + 1). Calculate old sum from old mean and count.

Question 19

Correct Answer: D) 1/11

$P(\text{both blue}) = (4/12)(3/11) = 12/132 = 1/11$.

Tip: Without replacement: first draw 4/12, second draw 3/11. Multiply the probabilities.

Question 20

Correct Answer: 15

$P = 500 \cdot 2^{(t/3)}$. We need $500 \cdot 2^{(t/3)} > 10,000$, so $2^{(t/3)} > 20$. Since $2^4 = 16 < 20$ and $2^5 = 32 > 20$, $t/3$ must exceed 4, so $t > 12$. At $t = 15$: $500 \cdot 2^5 = 16,000 > 10,000$. The answer is 15 hours.

Tip: Set up the inequality and find the smallest whole number of hours. Since doubling happens every 3 hours, t must be a multiple of 3.

Question 21

Correct Answer: A) 5/13

In right triangle PQR with right angle at Q: $\sin(P) = \text{opposite/hypotenuse} = QR/PR = 5/13$. $\cos(R) = \text{adjacent/hypotenuse} = QR/PR = 5/13$. So $\cos(R) = \sin(P) = 5/13$.

Tip: In a right triangle, sin of one acute angle equals cos of the other. $\sin(P) = \cos(R)$ when $P + R = 90^\circ$.

Question 22

Correct Answer: 22/3

$|2x - 3| = |x + 5|$. Case 1: $2x - 3 = x + 5$, so $x = 8$. Case 2: $2x - 3 = -(x + 5) = -x - 5$, so $3x = -2$, $x = -2/3$. Both solutions check out. Sum = $8 + (-2/3) = 24/3 - 2/3 = 22/3$.

Tip: Remove absolute values by considering both cases: the expressions are equal, or they are negatives of each other. Check both solutions in the original equation.

End of Answer Key

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